بِسۡمِ ٱللَّهِ ٱلرَّحۡمَٰنِ ٱلرَّحِيمِ

**In the name of Allah, Most Gracious, Most Merciful**

# CSE 4303
# Data Structure

Topic: Self Balancing Tree (AVL) Tree

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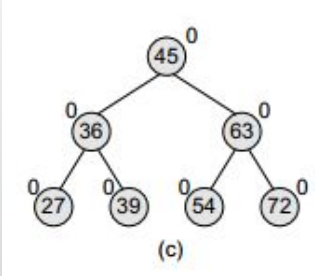
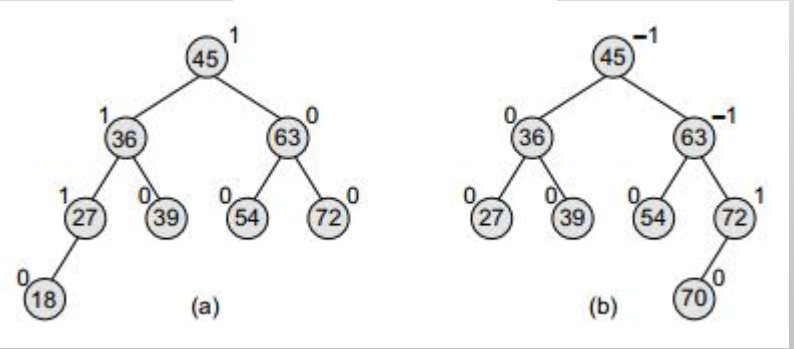
# Adelson-Velsky and E.M. Landis Tree (AVL)

Self-balancing binary search tree invented by G.M. Adelson-Velsky and E.M. Landis in 1962. Also known as a height-balanced tree.

- The heights of the two sub-trees of a node may differ by at most one.
- The height of the tree is limited to O(log n).
- An additional variable called the BalanceFactor.




Balance factor

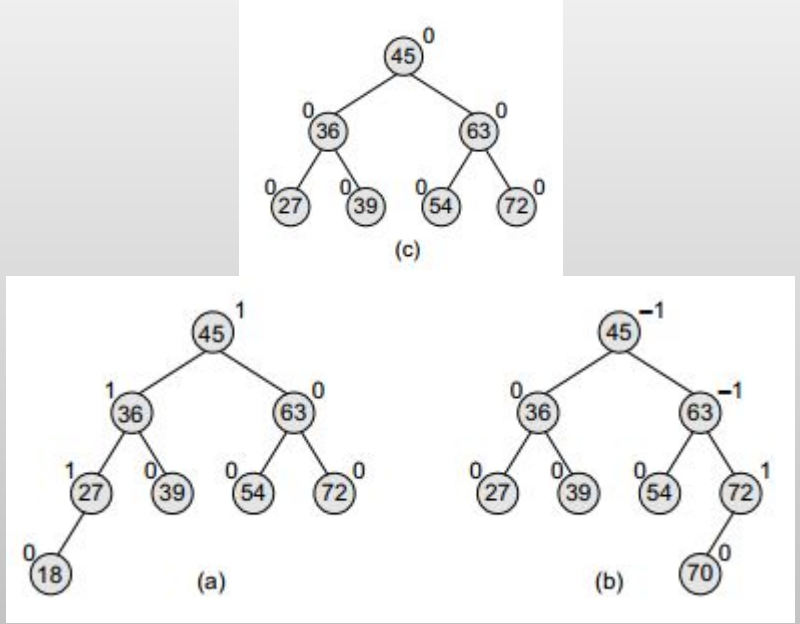
= Height (left sub-tree) – Height (right sub-tree)

Since the height of the tree is limited to O(log(n)), the time needed to search an element in the tree is greatly reduced.

# AVL Tree (Balance Factor)

***Balance factor = Height (left sub-tree) – Height (right sub-tree)***

- **Left-heavy tree:** If the balance factor of a node is 1, then it means that the left sub-tree of the tree is one level higher than that of the right sub-tree.
- **Perfectly balanced Tree:** If the balance factor of a node is 0, then it means that the height of the left sub-tree (longest path in the left sub-tree) is equal to the height of the right sub-tree.
- **Right-heavy tree:** If the balance factor of a node is –1, then it means that the left sub-tree of the tree is one level lower than that of the right sub-tree.



# Operations on AvL Trees

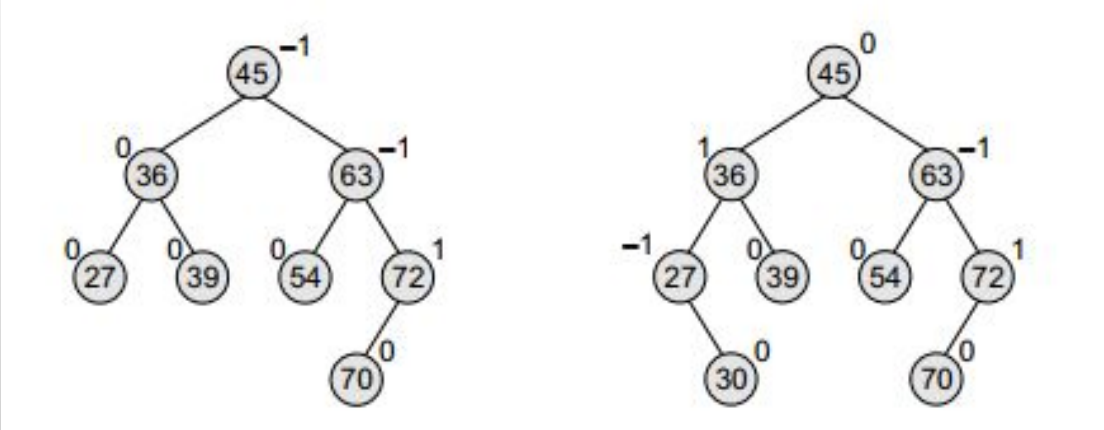
- Searching for a Node in an AVL Tree
  - Exactly the same way as it is performed in a binary search tree.
  - The search operation takes O(log n)
- Inserting a New Node in an AVL Tree
  - Done in the same way as it is done in a binary search tree.
  - The new node is always inserted as the leaf node.
  - Rotation is done to restore the balance of the tree.
  - If balance factor is in between -1 and 1 than no rotation is required
- Deleting a Node from an AVL Tree
  - Is similar to that of binary search trees.
  - Deletion may disturb the balance of the tree
  - Rotation is done to rebalance the tree if tree loses the AVLness

# Insertion in AvL Trees

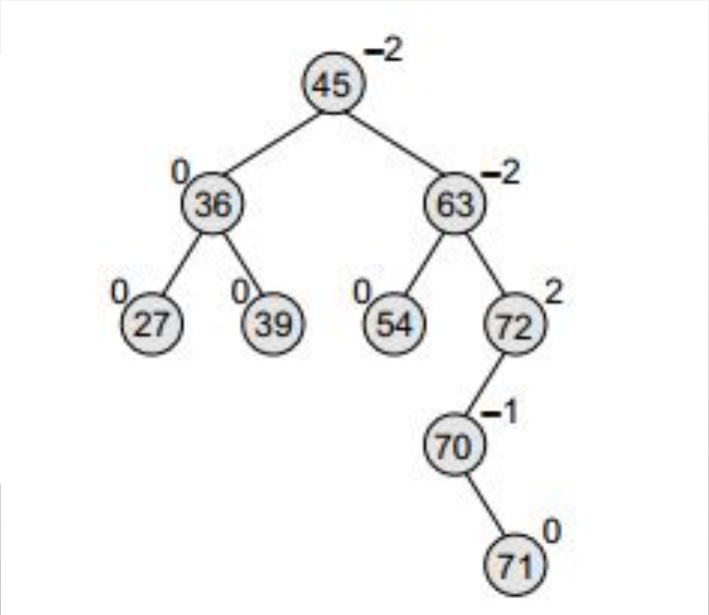
- The balance factor of the newly inserted node is always zero.
- But nodes on path from root to newly inserted node may get unbalanced by balance factor.
- 3 cases in insertion:
  - Initially, the node was either left- or right-heavy and after insertion, it becomes balanced.
  - Initially, the node was balanced and after insertion, it becomes either left- or right-heavy.
  - Initially, the node was heavy (either left or right) and the new node has been inserted in the heavy sub-tree, thereby creating an unbalanced sub-tree.

**Critical Node:** The unbalanced node after insertion.

# Insertion in AvL Trees




Insertion of 30 with no balancing requirement.




Insertion of 71 with balancing requirement.

# Insertion in AvL Trees

The four categories of rotations are:

- **LL rotation** The new node is inserted in the left sub-tree of the left sub-tree of the critical node.
- **RR rotation** The new node is inserted in the right sub-tree of the right sub-tree of the critical node.
- **LR rotation** The new node is inserted in the right sub-tree of the left sub-tree of the critical node.
- **RL rotation** The new node is inserted in the left sub-tree of the right sub-tree of the critical node

# Insertion in AvL Trees (LL Rotation)

While rotation, node B becomes the root, with T1 and A as its left and right child. T2 and T3 become the left and right sub-trees of A.

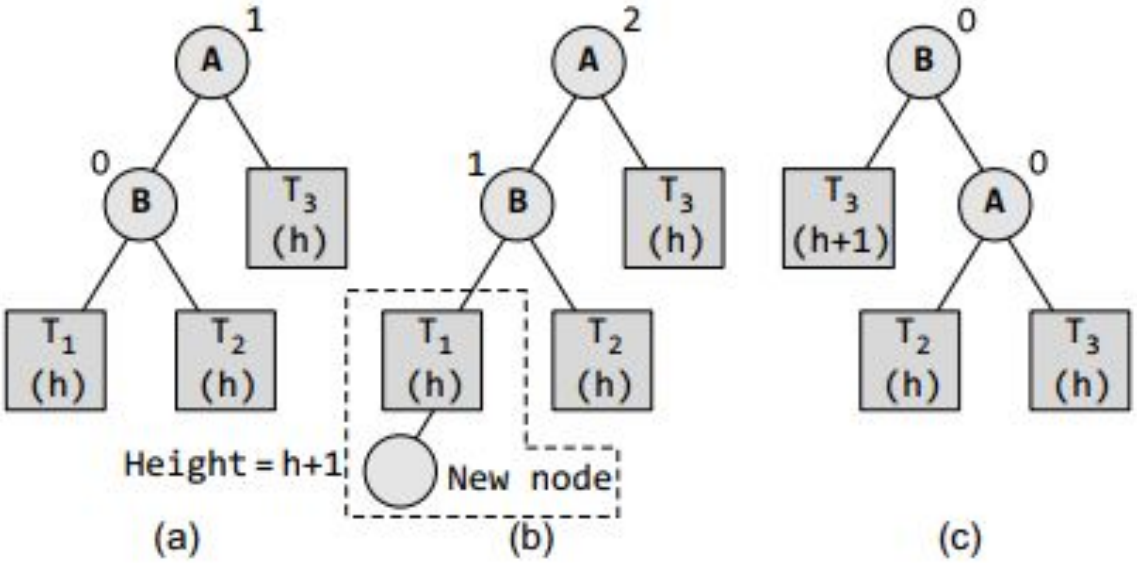



**Figure 10.40** LL rotation in an AVL tree

# Insertion in AvL Trees (LL Rotation)

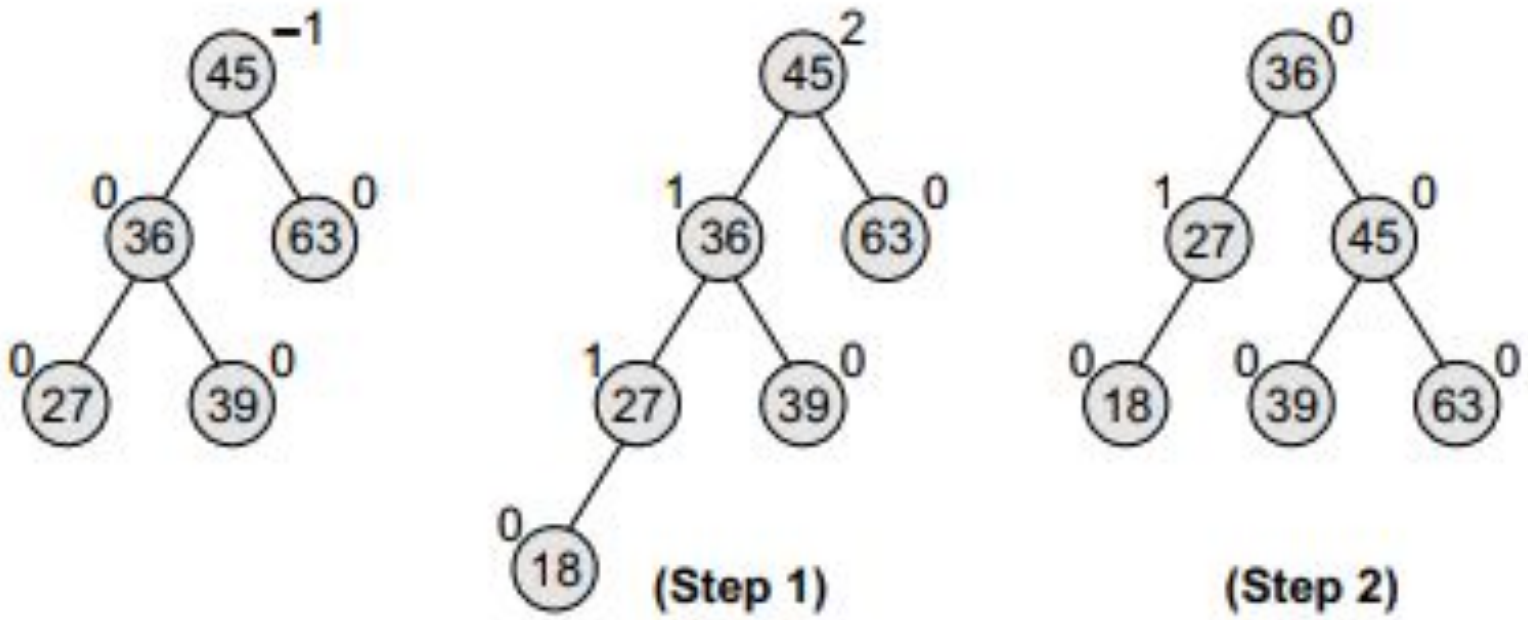



Figure 10.41 AVL tree

# Insertion in AvL Trees (RR Rotation)

While rotation, node B becomes the root, with A and T 3 as its left and right child. T1 and T2 become the left and right sub-trees of A.

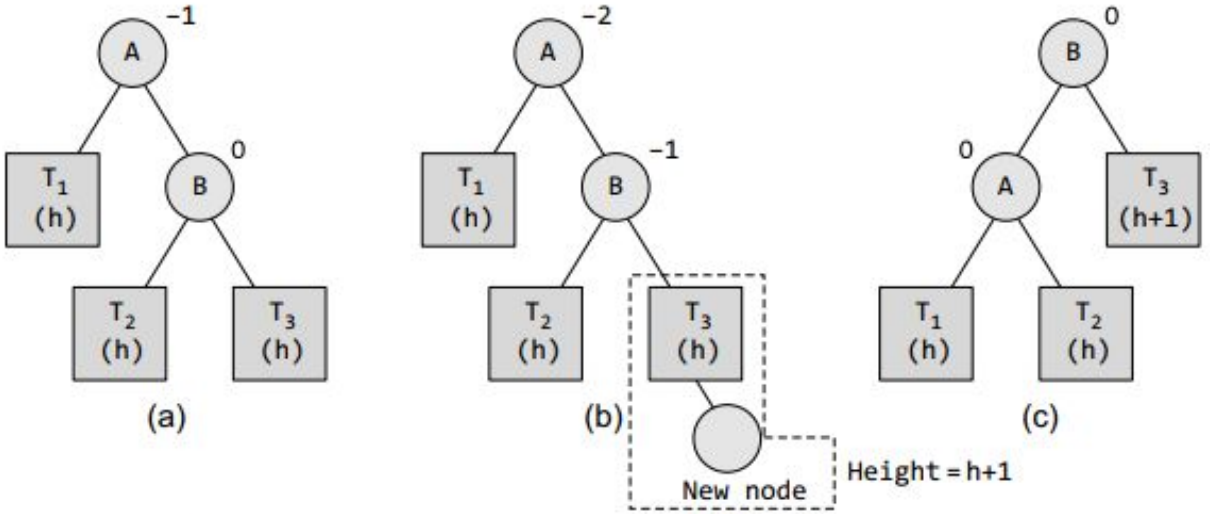
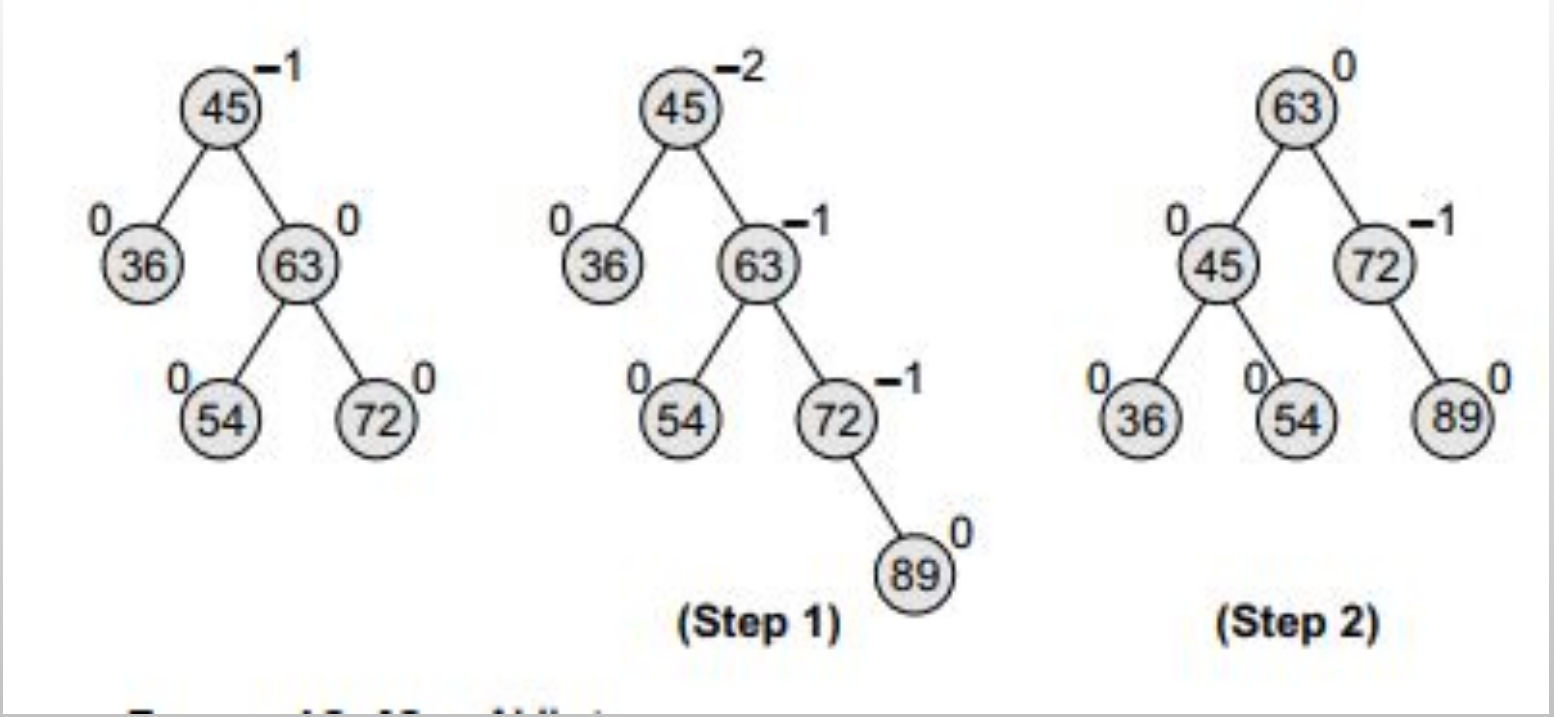



**Figure 10.42** RR rotation in an AVL tree

# Insertion in AvL Trees (RR Rotation)



# Insertion in AvL Trees (LR Rotation)

While rotation, node C becomes the root, with B and A as its left and right children. Node B has T1 and T2 as its left and right sub-trees and T3 and T 4 become the left and right sub-trees of node A

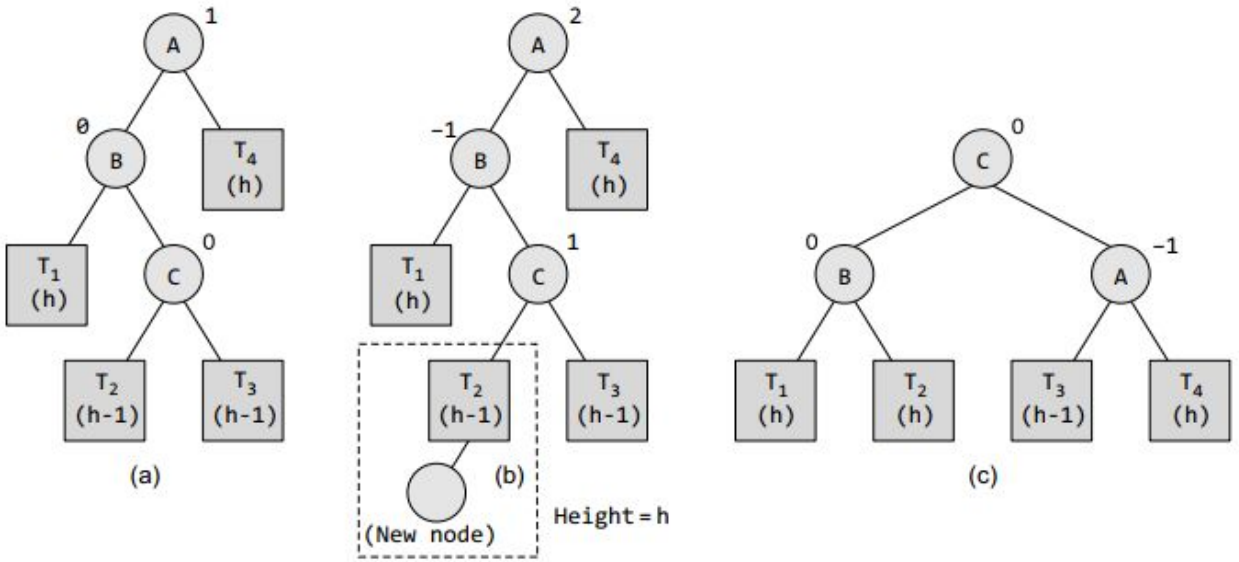
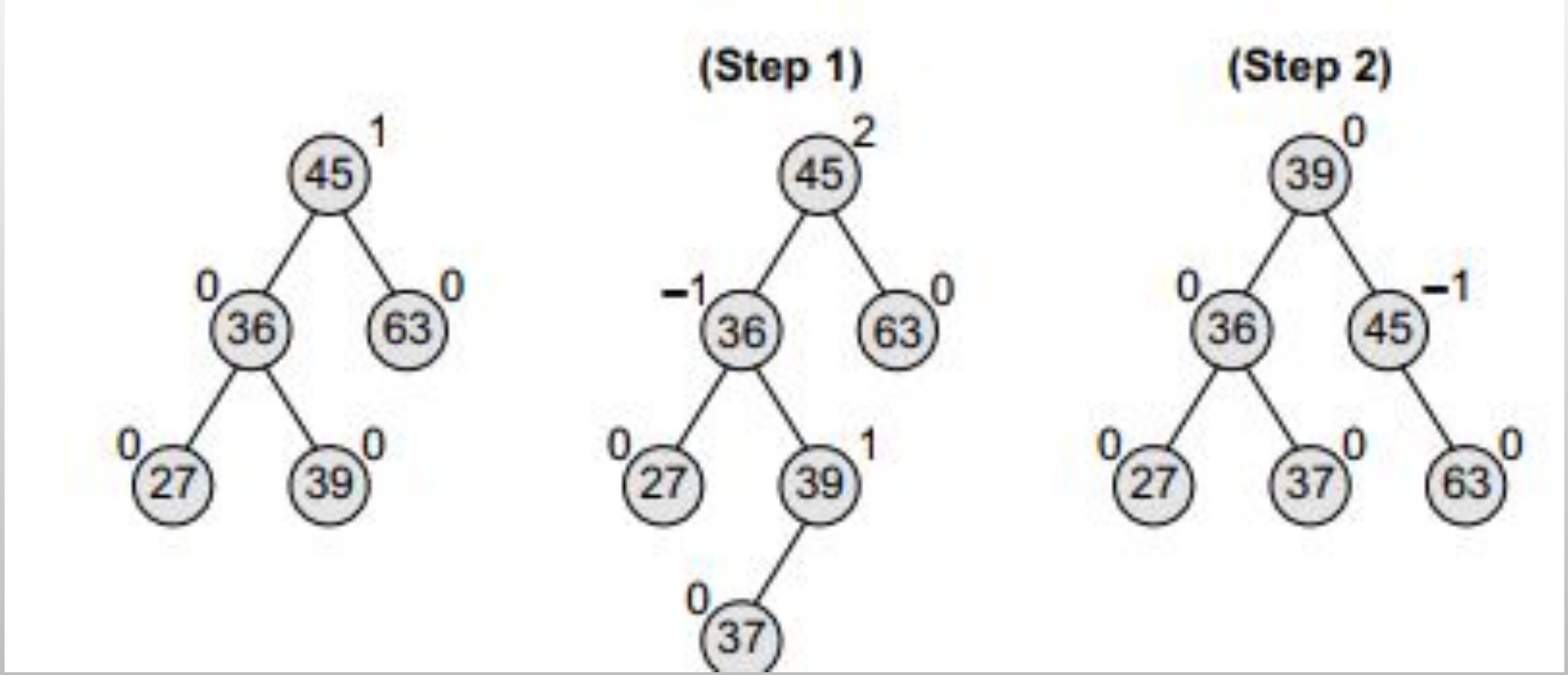


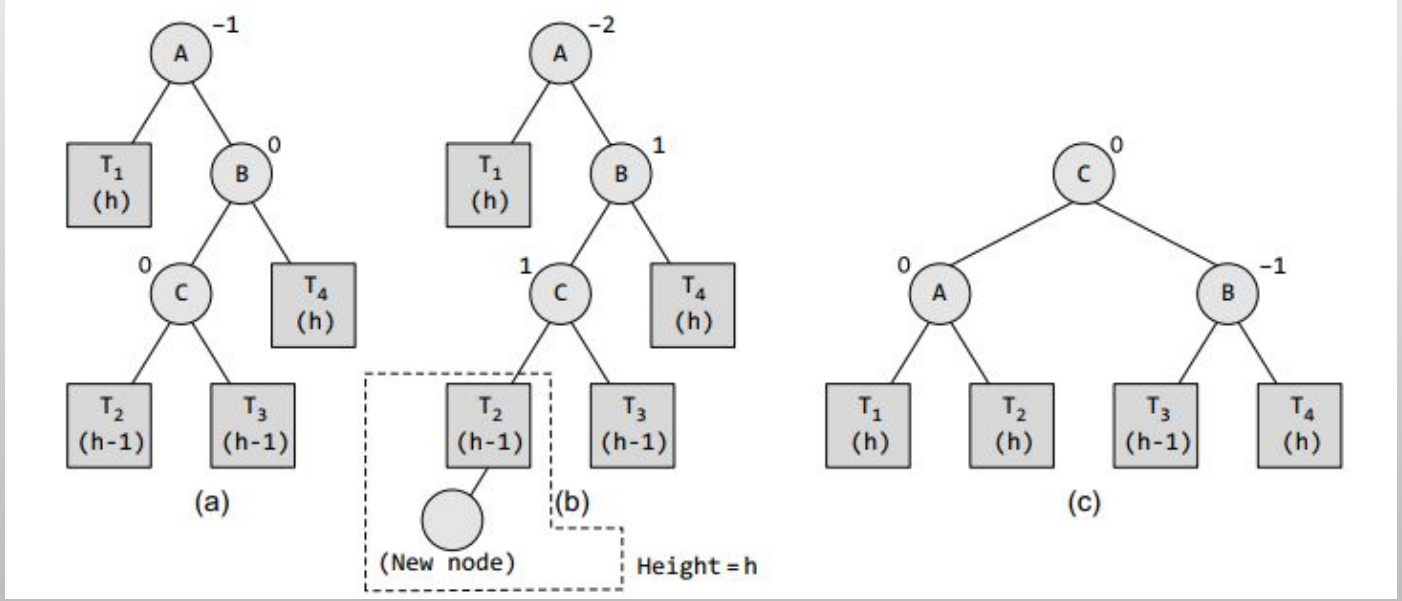

**Figure 10.44** LR rotation in an AVL tree

# Insertion in AvL Trees (LR Rotation)



# Insertion in AvL Trees (RL Rotation)

While rotation, node C becomes the root, with A and B as its left and right children. Node A has T1 and T2 as its left and right sub-trees and T3 and T4 become the left and right sub-trees of node B.



**Example 10.6** Construct an AVL tree by inserting the following elements in the given order. 63, 9, 19, 27, 18, 108, 99, 81.

***Solution***

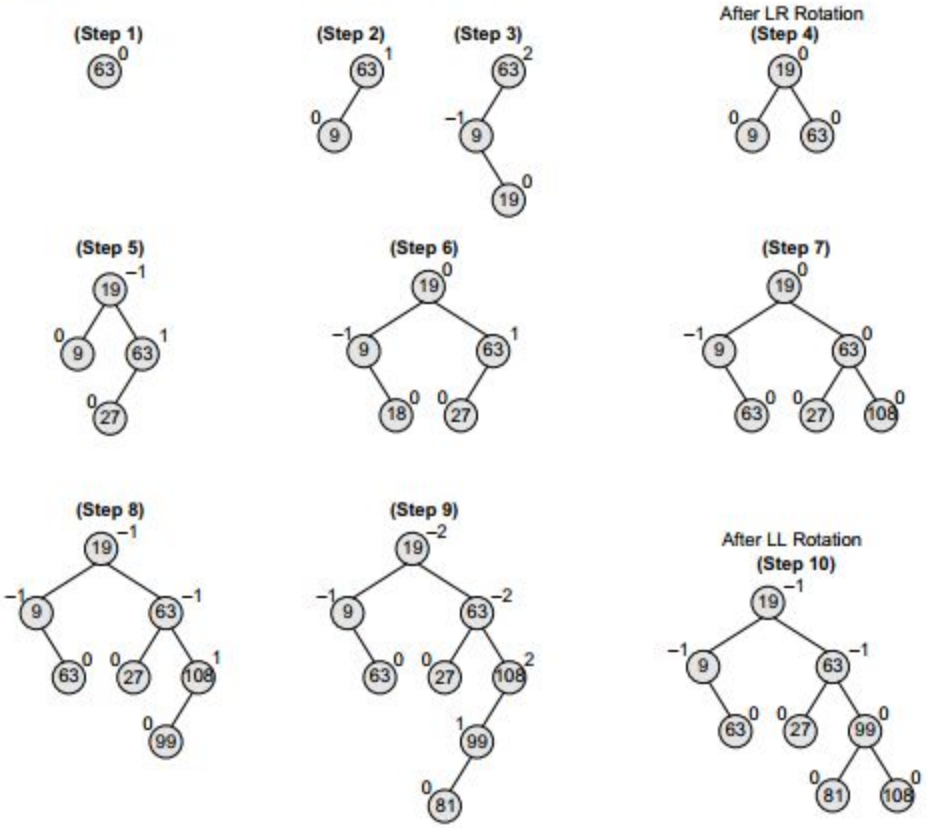
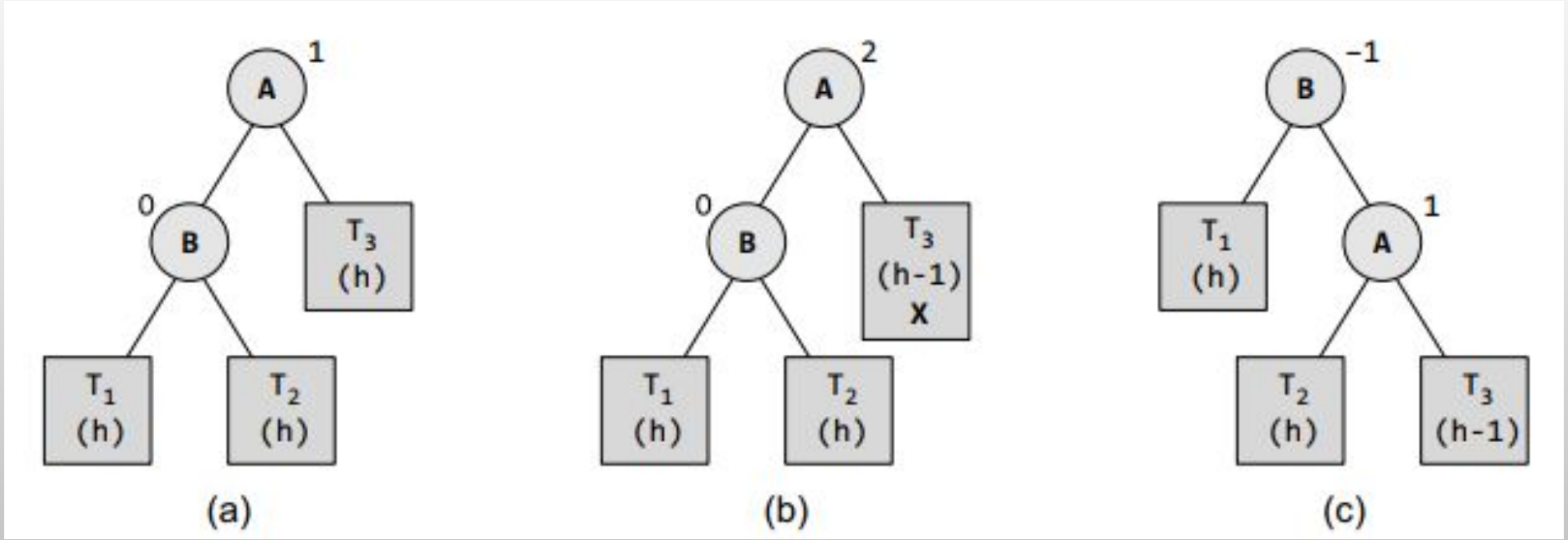
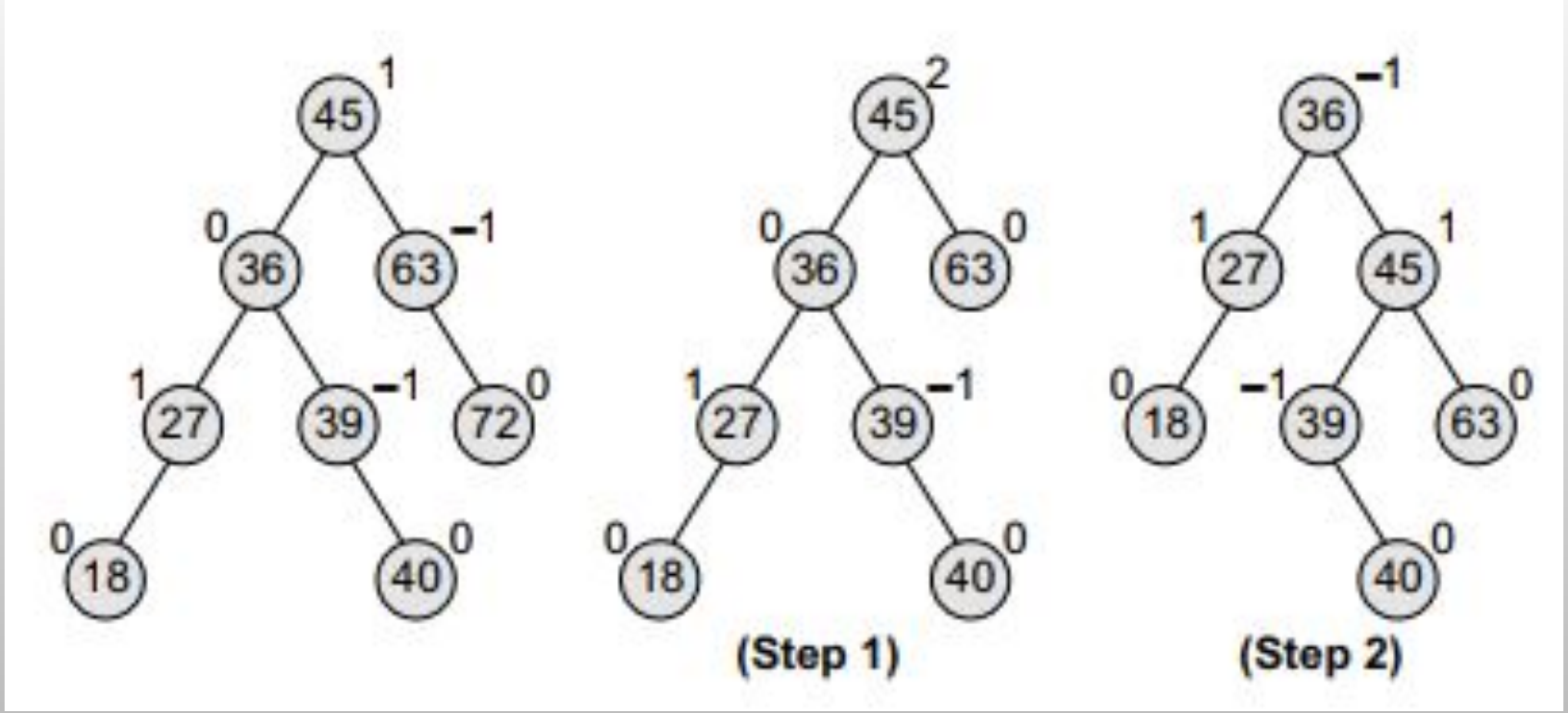



**Figure 10.47** AVL tree

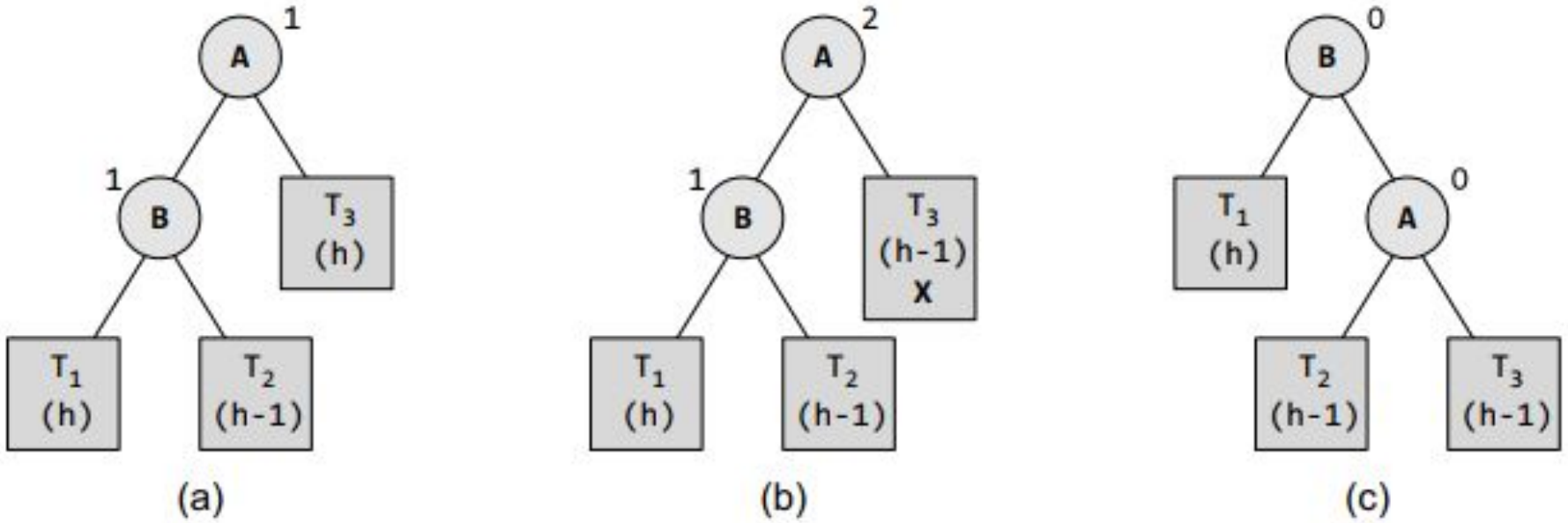
# Deletion in AvL Trees (R0 Rotation)



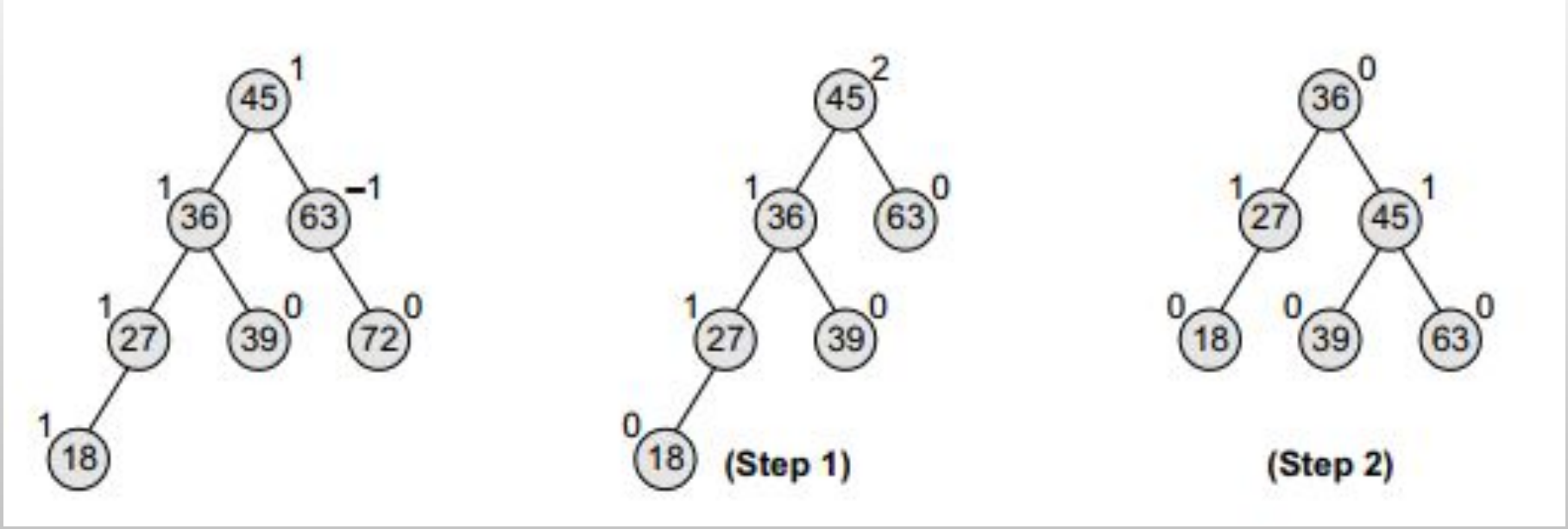
# Deletion in AvL Trees (R0 Rotation)



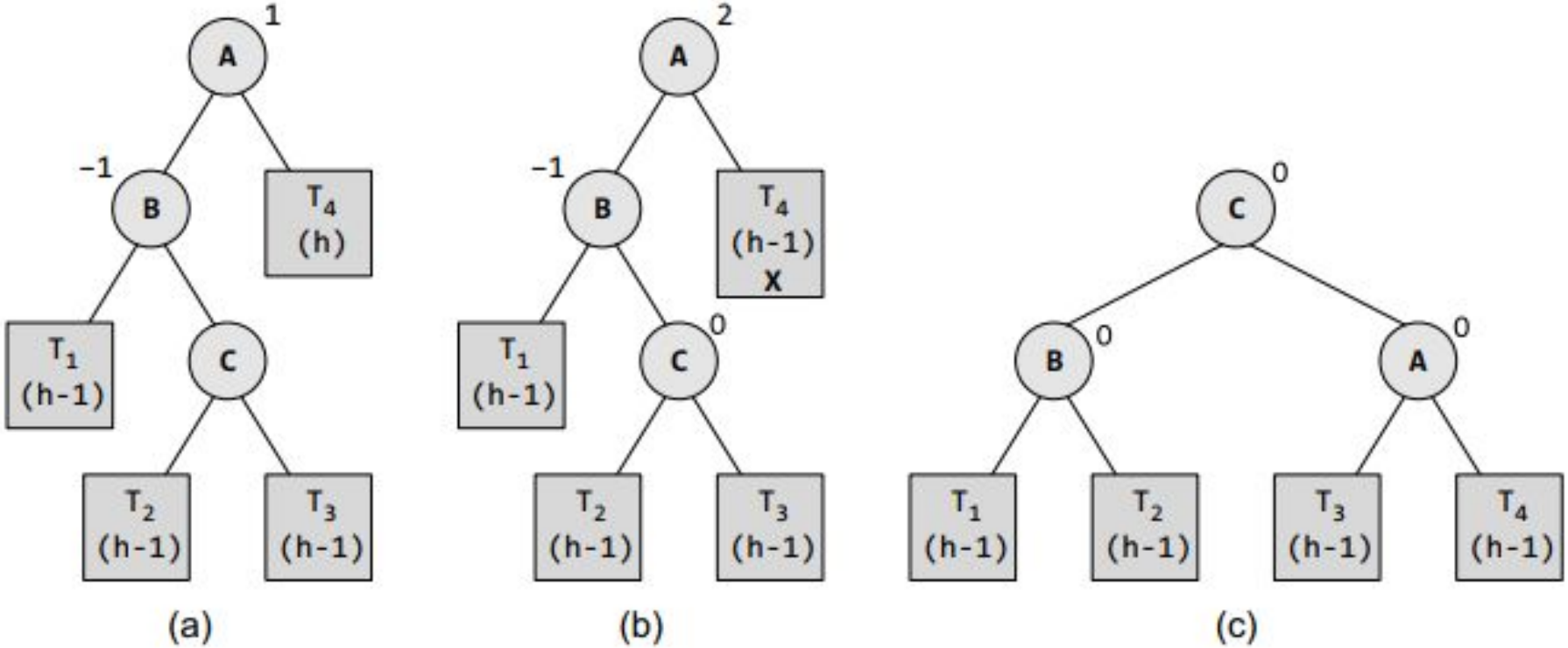
# Deletion in AvL Trees (R1 Rotation)



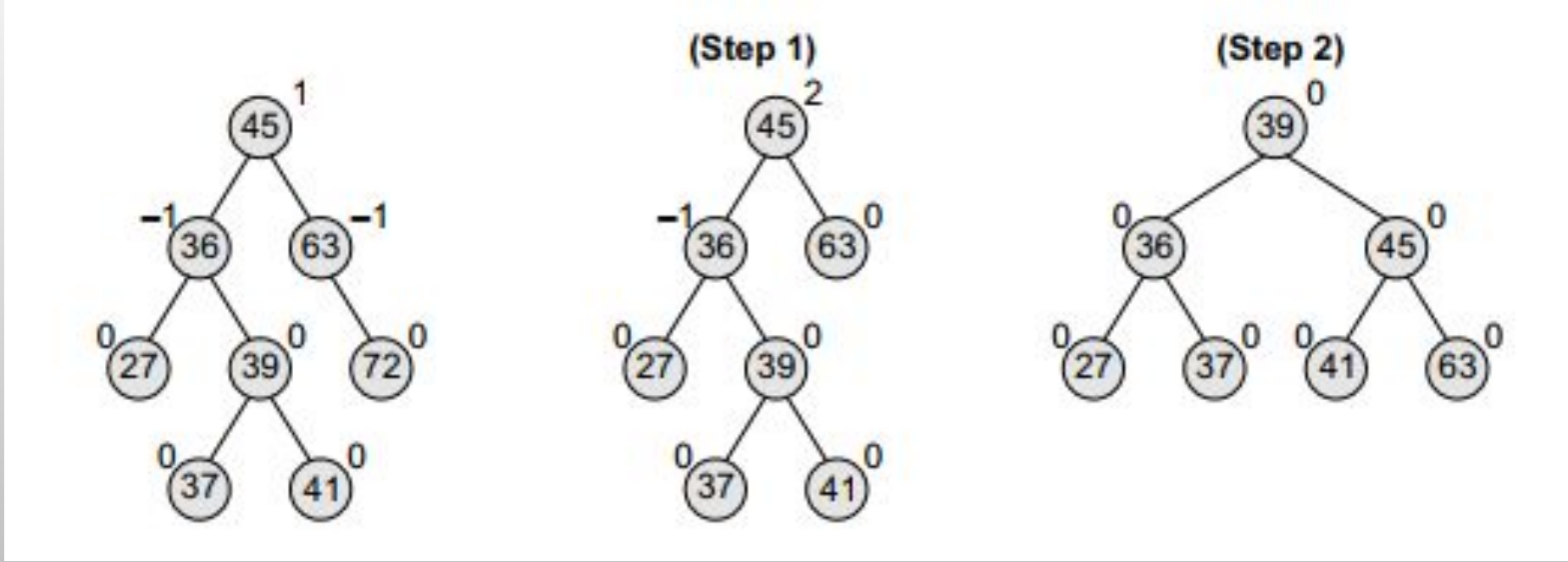
# Deletion in AvL Trees (R1 Rotation)



# Deletion in AvL Trees (R-1 Rotation)

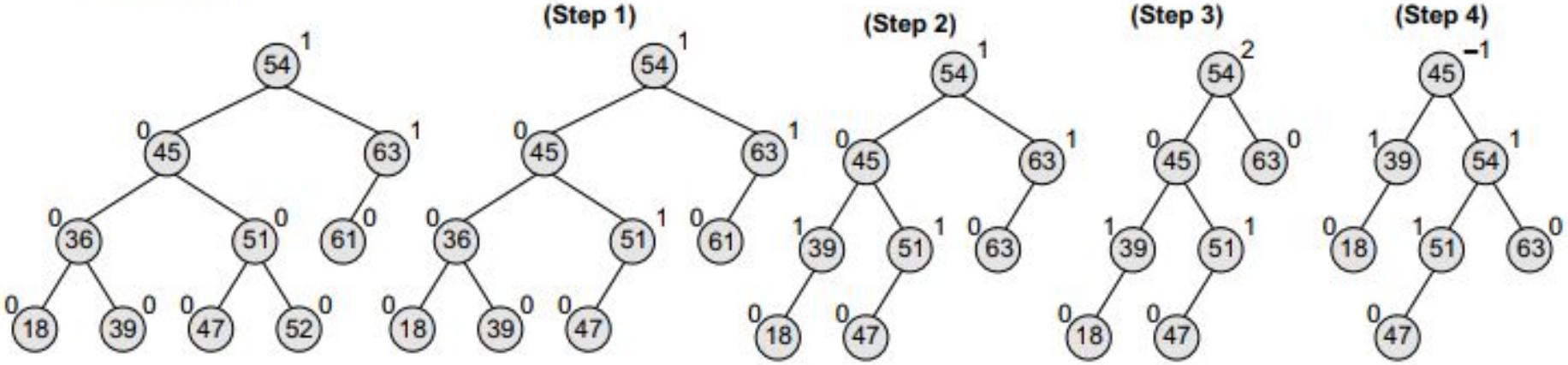


# Deletion in AvL Trees (R-1 Rotation)



**Example 10.10** Delete nodes 52, 36, and 61 from the AVL tree given in Fig. 10.54.

***Solution***



**Acknowledgements**

Data Structures
Using
c
Reema Thareja